Supplementary Material: StateShift: A Confirmatory Study of Reasoning-Context Continuation and Forced Restart in Mathematical Language Models

Sham Satish Thakare
Independent AI Researcher
github.com/shamddd

October 2026

1 Strict Contamination Sensitivity Analysis (Study 1)

Under strict decontamination filtering ($N_{\text{Strict}} = 388$), removing any problem with n-gram overlap ($n \geq 13$) or high embedding cosine similarity (> 0.85) against training sets, the state-by-checkpoint interaction remains virtually identical:

$$\Gamma_{256,\text{Strict}} = +0.1160 \quad (95\% \text{ CI } [+0.0913, +0.1408], p < 10^{-18}). \tag{1}$$

2 Nine-Checkpoint Empirical Contrast Table (Study 1)

Table 1 details the exact cell means and interaction estimates across all nine sampled checkpoints along the DeepScaleR-7B lineage ($N = 454$ problem pairs, $29,056$ row-level outcome records).

Table 1: Complete Nine-Checkpoint Empirical Interaction Contrast Table.

Checkpoint (t)	$\mu_{R,t}$	$\mu_{C,t}$	$\Delta_{R,t}$	$\Delta_{C,t}$	Γ_t	95% Bootstrap CI
0	0.3834	0.3892	0.0000	0.0000	+0.0000	—
32	0.4086	0.3811	+0.0252	-0.0081	+0.0333	[+0.0023, +0.0677]
64	0.4537	0.4258	+0.0703	+0.0366	+0.0337	[+0.0081, +0.0610]
96	0.5044	0.4328	+0.1210	+0.0436	+0.0774	[+0.0390, +0.1116]
128	0.5609	0.4919	+0.1775	+0.1027	+0.0748	[+0.0462, +0.1025]
160	0.5958	0.5419	+0.2124	+0.1527	+0.0598	[+0.0238, +0.0935]
192	0.6571	0.5653	+0.2737	+0.1761	+0.0976	[+0.0695, +0.1252]
224	0.6641	0.5749	+0.2807	+0.1857	+0.0950	[+0.0627, +0.1282]
256	0.7039	0.5921	+0.3205	+0.2029	+0.1176	[+0.0955, +0.1400]

3 Exact Discrete Binomial Power Modeling (Study 2)

In paired binary trials, let $K \sim \text{Binomial}(N, q)$ denote the number of discordant pairs, where $q = p_{10} + p_{01}$. Conditional on K, the number of successes under Condition F follows $B \sim \text{Binomial}(K, \theta)$,

where $\theta = p_{10}/q$. Under the null hypothesis $H_0 : \theta = 0.50$, the exact two-sided rejection region at level $\alpha = 0.05$ is defined by critical values $c_L(K)$ and $c_U(K)$ such that $P_{H_0}(B \leq c_L \text{ or } B \geq c_U) \leq 0.05$.

The unconditional power of the exact McNemar test is the expectation over the distribution of K:

$$\text{Power}(N, q, \theta) = \sum_{k=1}^{N} \binom{N}{k} q^k (1-q)^{N-k} \left[\sum_{b=0}^{c_L(k)} \binom{k}{b} \theta^b (1-\theta)^{k-b} + \sum_{b=c_U(k)}^{k} \binom{k}{b} \theta^b (1-\theta)^{k-b} \right] \quad (2)$$

Under pilot parameters ($q = 0.10, \theta = 8/10 = 0.80$), unconditional power is 36.19% at $N = 100$. The minimum sample size achieving $\geq 80.0\%$ unconditional power is $N = 231$ paired problems ($1 - \beta = 80.02\%$).

4 Confirmatory Hardware and Pinned Revisions (Study 3)

All 924 confirmatory rollouts were executed on RunPod Secure Cloud:

- **Accelerator**: NVIDIA GeForce RTX 4090 (24,564 MiB VRAM, Driver 595.91.07, CUDA 13.0).
- **Model**: `Qwen/Qwen2.5-Math-7B-Instruct` pinned to commit hash `ef9926d75ab1d54532f6a30dd5e760355e`
- **Decoding**: Deterministic greedy search ($\tau = 0.0$, seed $= 42$, max new tokens $= 768$).
- **Financial Spend**: $0.5795 USD active compute charges (38.15 minutes).
- **Teardown**: Verified via authenticated GraphQL query returning 0 active pods and 0 active volumes.

5 Complete Listing of the 23 Discordant Problems

Table 2 provides the complete accounting of all 23 discordant problems between Conditions D and F in the confirmatory trial.

Table 2: Complete Listing of the 23 Discordant Problems ($N = 231$, Study 3).

Cell	Problem ID	Subject	Level	Gold Answer	Condition D Outcome	Condition F Outcome
b	`math500_138`	Counting & Prob	4	$1/3$	$1/6$ (`stop`, 288 tok)	$1/3$ (`stop`, 319 tok) [Correct]
b	`math500_156`	Geometry	4	30°	Truncated (`length`, 768 tok)	30 (`stop`, 696 tok) [Correct]
b	`math500_173`	Algebra	3	4	Truncated (`length`, 768 tok)	4 (`stop`, 649 tok) [Correct]
b	`math500_231`	Algebra	3	10	Truncated (`length`, 768 tok)	10 (`stop`, 475 tok) [Correct]
b	`math500_296`	Intermediate Alg	5	(B)	Truncated (`length`, 768 tok)	B (`stop`, 381 tok) [Correct]
b	`math500_303`	Algebra	3	28	33.6 (`stop`, 609 tok)	28 (`stop`, 340 tok) [Correct]
b	`math500_326`	Precalculus	3	$9/100$	Truncated (`length`, 768 tok)	$9/100$ (`stop`, 632 tok) [Correct]
b	`math500_376`	Geometry	1	$(8, -2)$	$(22/3, -4/3)$ (`stop`, 722 tok)	$(8, -2)$ (`stop`, 562 tok) [Correct]
b	`math500_400`	Geometry	3	$14/3$	Truncated (`length`, 768 tok)	$14/3$ (`stop`, 631 tok) [Correct]
b	`math500_435`	Intermediate Alg	1	$(-\infty, 2) \cup (3, \infty)$	Truncated (`length`, 768 tok)	$(-\infty, 2) \cup (3, \infty)$ (`stop`, 749 tok) [Correct]
b	`math500_466`	Precalculus	5	8π	Truncated (`length`, 768 tok)	8π (`stop`, 608 tok) [Correct]
c	`math500_139`	Intermediate Alg	2	1	1 (`stop`, 432 tok) [Correct]	Truncated (`length`, 768 tok)
c	`math500_163`	Intermediate Alg	4	$-35/9$	$-35/9$ (`stop`, 518 tok) [Correct]	$-31/9$ (`stop`, 532 tok)
c	`math500_219`	Precalculus	4	$1/2$	$1/2$ (`stop`, 449 tok) [Correct]	Truncated (`length`, 768 tok)
c	`math500_252`	Prealgebra	2	81	81 (`stop`, 627 tok) [Correct]	77 (`stop`, 655 tok)
c	`math500_272`	Prealgebra	1	1030	1030 (`stop`, 745 tok) [Correct]	Truncated (`length`, 768 tok)
c	`math500_304`	Intermediate Alg	4	9	9 (`stop`, 690 tok) [Correct]	Truncated (`length`, 768 tok)
c	`math500_385`	Number Theory	1	22	22 (`stop`, 498 tok) [Correct]	3 (`stop`, 451 tok)
c	`math500_398`	Counting & Prob	5	Navin	Navin (`stop`, 484 tok) [Correct]	Ian (`stop`, 470 tok)
c	`math500_403`	Number Theory	4	3	3 (`length`, 768 tok) [Correct]	4 (`stop`, 585 tok)
c	`math500_409`	Precalculus	1	$2\sqrt{5}$	$2\sqrt{5}$ (`stop`, 715 tok) [Correct]	10 (`length`, 768 tok)
c	`math500_416`	Precalculus	4	$1/8$	$1/8$ (`stop`, 569 tok) [Correct]	Truncated (`length`, 768 tok)
c	`math500_462`	Number Theory	2	59	59 (`stop`, 509 tok) [Correct]	Truncated (`length`, 768 tok)